

MIS772 – Predictive Analytics

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Assignment 2 – Individual
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Executive summary

Executive problem statement:

Airbnb is eager to enhance their understanding of customer satisfaction within the Madrid rental market by analysing customer review data. Airbnb aims to leverage the company's extensive property listings and customer review data to identify key drivers of guest satisfaction. By developing sophisticated predictive models, Airbnb seeks to forecast review scores using key property features and guest feedback metrics.

This data-driven approach will empower Airbnb and its property partners to make informed decisions that enhance the overall guest experience. The predictive model will serve as a powerful tool for hosts, enabling them to proactively address potential areas of improvement and align their offerings more closely with guest expectations.

Executive solution statement:

The analysis shows that there is a slight positive relationship of positive customer reviews and positive host descriptions, indicating that customers focus more on the actual experience rather than online description.

The analysis has also identified key factors that influence customer satisfaction, specifically measured by the review scores given by Airbnb guests, such as accuracy, cleanliness, check-in experience, communication, location and value. The results showed that:

- Accuracy and value have the strongest relationship of high ratings, with improvements in these areas having a significant positive effect on the overall review score.
- Cleanliness, check-in experience and communication also played significant roles in increasing customer satisfaction.
- Location had a minor negative impact, possibly due to trade-offs customers might make in high-demand areas, such as paying higher prices or compromising on certain amenities.

To enhance guest satisfaction and further improve review scores, Airbnb should focus on the following strategic actions:

- Hosts should ensure their property descriptions and amenities accurately reflect what is being offered.
- Improve cleaning and communication standards.
- Adjust price for high-demand but less convenient locations.

By implementing these solutions, Airbnb will be able to:

- Increase customer satisfaction by addressing key factors that impact review scores.
- Enable hosts to improve the accuracy of their listings, maintain cleanliness, and provide better communication, leading to higher ratings and increased bookings.
- Adjust pricing strategies based on location insights, improving guest perception of value.
- Utilize advanced predictive modeling to offer more accurate forecasts of guest satisfaction, ultimately improving decision-making and strategic planning.

These actions will help Airbnb maintain its competitive advantage, improve guest experiences, and foster stronger relationships with property hosts.

A. Sentiment Score Correlation

The objective is to analyse correlation between hosts' and customers' sentiment score (calculated by positive words minus negative words) for Airbnb properties.

The relevant data attributes are: hosts' description from listing file and customers' comment from reviews file.

Data Process:

- 'Filter Examples' is used to remove 1138 missing descriptions.
- Specific attributes from the data are selected for further analysis. In this case, ID and description attributes are chosen in 'Select Attributes'.
- 'Nominal to Text' operator is used to convert the data type into text format for comparison.
- For further investigation, ID attribute is set as ID in 'Set Role'.
- Description attribute, positive and negatives words are processed for sentiment analysis as below:
 - Transform cases (converts all text into lowercase) → tokenize (breaks down the text into words) → stem (reduce words to their root form) → filter stopwords (remove common stopwords) → filter tokens (retain 4–25-character words).

Processed description is compared with positive and negative words to quantify the sentiment.

- 'Generate Aggregation' and 'Aggregate' operators count the total number of positive and negative words.
- The datasets are joined together on ID, combining the positive and negative score by 'Join'.
- Host's sentiment score is generated by 'Generate Attributes' (positive minus negative).

The same process is applied for customers (reviews file), with comments attribute used to calculate the positive and negative score and listing_id attribute used to join the data.

After obtaining both sentiment scores from host and customer, the data is combined using id from host dataset and listing_id from customer dataset. 'Correlation Matrix' is used to examine the correlation as shown in Figure 1.

Attributes	sum(negcount_h)	sum(poscount_h)	sentiment (host)	sum(negcount_c)	sum(poscount_c)	sentiment (customer)
sum(negcount_h)	1	0.782	0.534	0.205	0.195	0.190
sum(poscount_h)	0.782	1	0.945	0.222	0.214	0.208
sentiment (host)	0.534	0.945	1	0.193	0.187	0.183
sum(negcount_c)	0.205	0.222	0.193	1	0.961	0.937
sum(poscount_c)	0.195	0.214	0.187	0.961	1	0.997
sentiment (customer)	0.190	0.208	0.183	0.937	0.997	1

Figure 1 - Hosts' raw sentiment score and customers' raw sentiment score correlation

Correlation coefficient between host's sentiment score and customer's sentiment score is 0.183, indicating a weak positive correlation. This shows a slight tendency for positive customer reviews to be associated with positive host descriptions. Other factors likely have a stronger influence on customer sentiment that need to be examined further.

B. Attributes Selection

The objective is to estimate the review_scores_rating, which represents the overall score given by customers, of Airbnb properties. Histogram shows that most properties score highly, clustering between 90-100, indicating that properties generally meet customers' expectations. It would be beneficial to explore other attributes' correlations with review_scores_rating to understand the ratings.

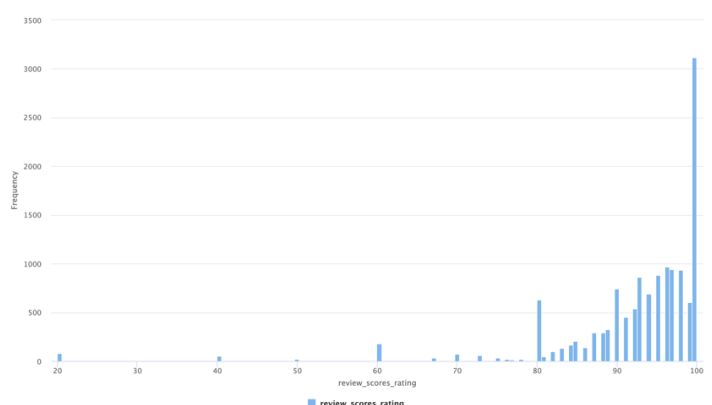


Figure 2 – Histogram of Review_scores_rating

The process of selecting predictors involves selecting appropriate data attributes, handling missing values, performing necessary transformations, and analysing data distribution and correlation as shown in Figure 3.

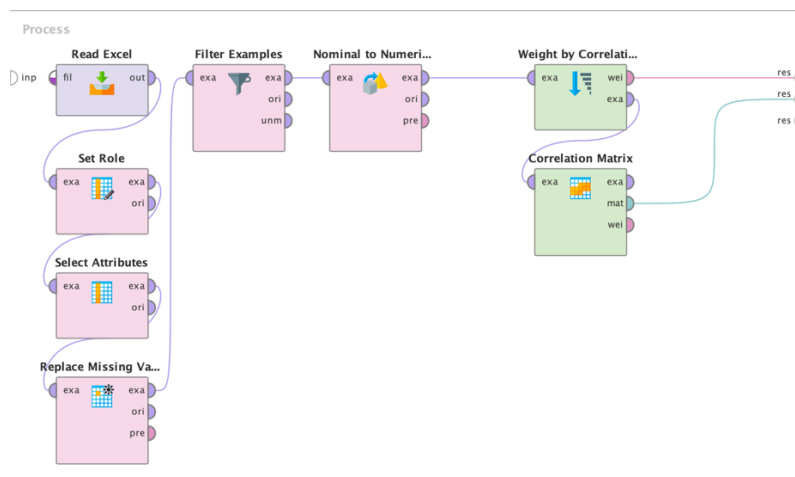


Figure 3 – Data selection process

- The following attributes are selected: price, accommodates, bedrooms, bathrooms_text, property_type, instant_bookable, review_scores_accuracy, review_scores_cleanliness, review_scores_communication, review_scores_checkin, review_scores_location, review_scores_value due to their potential impact on the target variable (review_scores_rating) by 'Select Attribute'.
- Review_scores_rating is set as label by 'Set Role'.
- Bedroom attribute's missing value is possibly due to studio apartments. 'Replace Missing Value' is used to replace missing values by 0. Other missing data is removed with 'Filter Examples' to improve dataset quality and model accuracy.
- 'Nominal to Numerical' to convert nominal data for correlation analysis.
- Correlation analysis is conducted using the 'Weight by Correlation' and 'Correlation Matrix'. The result is shown in Figure 4.

attribute	wei... ↓
review_scores_value	0.816
review_scores_accuracy	0.785
review_scores_cleanliness	0.736
review_scores_communication	0.695
review_scores_checkin	0.651
review_scores_location	0.474
instant_bookable	0.095
reviews_per_month	0.064
property_type	0.039
bedrooms	0.035
room_type	0.029
bathrooms_text	0.022
accommodates	0.021
neighbourhood_group_cleansed	0.021
price	0.007

Figure 4 – Correlation matrix ranked by weight

The analysis reveals that specific review scores (value, accuracy, cleanliness, communication, check-in, and location) have moderate to strong correlations with the overall rating. These factors are selected as predictors for our model. Other attributes show weak relationships with the overall rating and are excluded. Additionally, nominal attributes are omitted as they violate the linearity assumption required for the planned multiple linear regression model.

Predictive modelling

Multiple linear regression is selected as the primary estimation model because it is well-suited for predicting continuous variables like the review_scores_rating. The nature of the task involves estimating a numerical output, and linear regression is a simple yet effective method to model the relationship between independent predictor attributes and the dependent variable.

The process of building multiple linear regression model is shown in Figure 5.

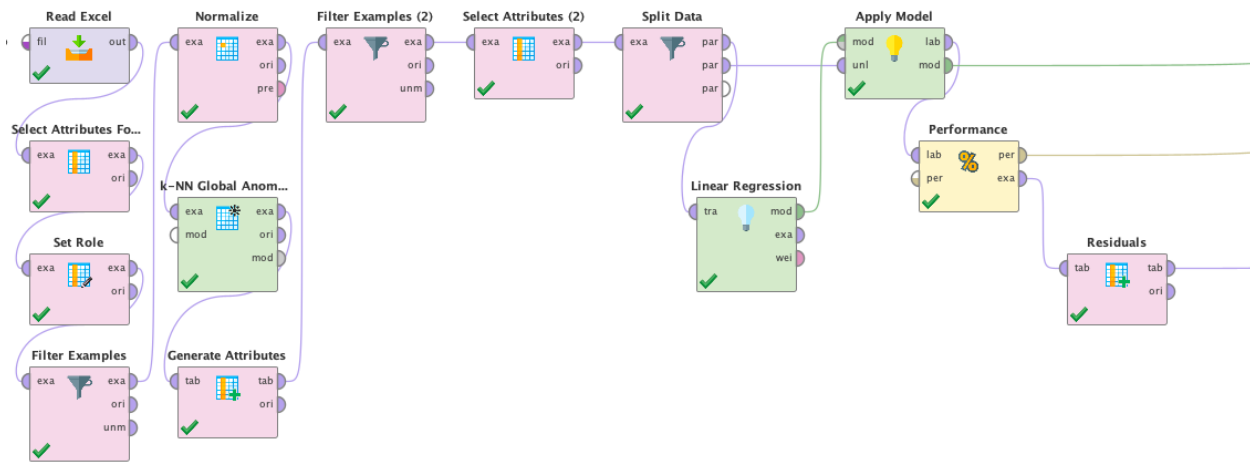


Figure 5 – Predictive modelling process

After loading the data, anomaly detection process is implemented to ensure multiple linear regression model assumption.

- All numerical attributes from listing file are chosen for outlier-checking using 'Select Attributes' with review_scores_rating is set as label by 'Set Role'.
- Missing values is removed using 'Filter Examples' to enhance data integrity.
- 'Normalize' is applied for all features with z-transformation method to convert all values into the same range to ensure that all attributes have equal influence, preventing attributes with larger scales from disproportionately influencing the model.
- k-NN Global Anomaly Score is deployed to detect anomalies in the dataset. The k value is set to 10, meaning that each data point's anomaly score is calculated based on its 10 nearest neighbors. "k = 10" as to ensure anomaly score is not biased by a very small or very large number of neighbors.
- The outlier range is between 0 to 9 as shown in Figure 6.

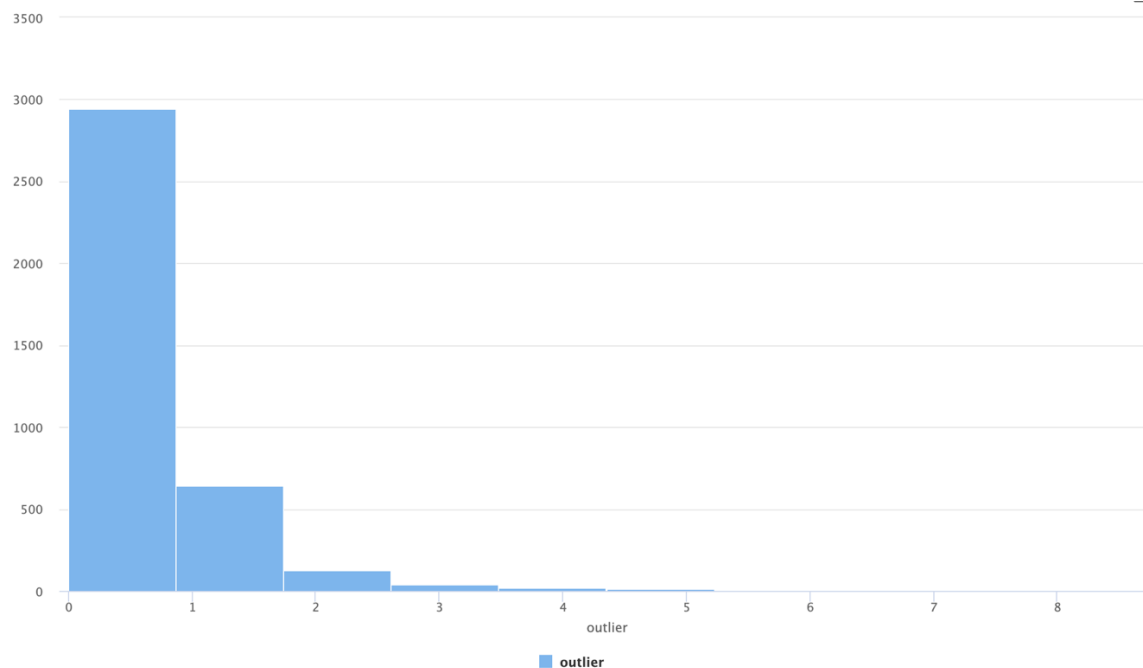


Figure 6 - Outliers

- To deal with extreme outliers, "Outlier Flag" attribute with function expression of outlier greater than 6 is generated by 'Generate Attribute'. After that, those values are removed by 'Filter Examples'.

Outliers can negatively impact the performance of regression models. This to ensure that the dataset is free of extreme outliers, allowing the model to be trained on clean data without extreme values that could distort the results and ensuring the assumption of normality of multiple linear regression model.

- After eliminating the extreme outliers, relevant attributes are chosen as model's predictors using 'Select Attributes'. As mentioned above, value, accuracy, cleanliness, communication, check-in and location review scores are chosen due to their strong correlation with the target variable. Other attributes are excluded because of their weak relationship with review_scores_rating and their assumption violation.
- The data is divided into two parts using 'Split Data' operator (0.7 and 0.3 ratio), allocating 70% for training and 30% for testing. This 70-30 split serves two purposes: the larger portion for training helps the model capture a wide range of patterns and relationships. While the smaller portion allows us to assess how well the model performs on unfamiliar data. This approach helps ensure the predictive model is both comprehensive in its learning and reliable in its predictions for new cases.
- 'Linear Regression' with 'M5 Prime' feature selection is applied, using a minimum tolerance threshold of 0.05 to ensure only features with a significant contribution are considered. The 'Apply Model' function is used to make predictions on a new dataset based on the previously trained model. For model evaluation, 'Performance' operator is deployed. Additionally, 'Residuals' attribute is generated, calculated as the difference between the observed and predicted values, to conduct further performance analysis.

The model is shown in Figure 7 as below:

Attribute	Coefficient	Std. Error	Std. Coefficient	Tolerance	t-Stat	p-Value
review_scores_accuracy	2.379	0.088	0.232	0.360	27.066	0
review_scores_cleanliness	1.986	0.076	0.197	0.452	26.044	0
review_scores_checkin	0.621	0.080	0.062	0.501	7.730	0.000
review_scores_communication	1.519	0.085	0.148	0.478	17.971	0
review_scores_location	-0.222	0.062	-0.022	0.720	-3.592	0.000
review_scores_value	3.943	0.084	0.390	0.381	47.078	0
(Intercept)	92.583	0.051	?	?	1832.982	0

Figure 7 – Multiple linear regression model

- The tolerance for all attribute is greater than 0.1, indicating that there is no significant multicollinearity between attributes. P-value for all attributes less than 0.05, indicating that they are all significant.
- review_scores_accuracy (Coefficient: 2.379) indicates that for every one-unit increase in the accuracy score, the review_scores_rating increases by 2.379 points. This means accuracy plays a significant role in customer satisfaction, and properties that match or exceed guest expectations in terms of accuracy see a notable boost in their overall rating.
- review_scores_cleanliness (Coefficient: 1.986) indicates that for every one-unit increase in the cleanliness score, the review_scores_rating increases by 1.986 points. Cleanliness is another highly valued factor, with guests giving higher ratings to properties that maintain a high standard of cleanliness.
- review_scores_checkin (Coefficient: 0.621) indicates that for every one-unit increase in the check-in score, the review_scores_rating increases by 0.621 points. While not as impactful as accuracy or cleanliness, ease of check-in still has a positive influence on the overall rating. A smooth check-in process contributes to guest satisfaction, but its effect is more moderate compared to other factors.
- review_scores_communication (Coefficient: 1.519) indicates that for every one-unit increase in the communication score, the review_scores_rating increases by 1.519 points. Good communication between the host and guests significantly boosts the overall rating. This shows that guests highly value responsiveness and clear communication from the host during their stay.
- review_scores_location (Coefficient: -0.222) indicates that for every one-unit increase in the location score, the review_scores_rating decreases by 0.222 points. This negative coefficient suggests that location, while important, may have a nuanced relationship with overall guest satisfaction. It is possible that properties in highly desirable locations may have other shortcomings, such as higher prices or overcrowding, that temper the positive impact of location on the overall rating.
- review_scores_value (Coefficient: 3.943) indicates that for every one-unit increase in the value score, the review_scores_rating increases by 3.943 points. Properties that are seen as providing good value for money receive the highest boost in overall ratings. This shows that offering good value is the most effective way for property owners to improve their ratings.

Model Evaluation

To evaluate the performance of the predictive model, root mean squared error (RMSE), squared correlation and residuals plot are used. The evaluation is demonstrated as below.

RMSE	Squared correlation
4.386	0.804

Figure 8 – Hold-out method evaluation metrics

- RMSE of 4.386 indicates that, on average, the model's predictions are 4.386 points away from the actual review_scores_rating. As the review_scores_rating original range is from 20-100, an RMSE of 4.386 is fairly reasonable, as it represents a relatively small error in the context of the overall range.
- Squared Correlation of 0.804 suggests that 80.4% of the variability in the review scores is explained by the model. This is a reasonably good fit, suggesting the model captures most of the key relationships in the data.

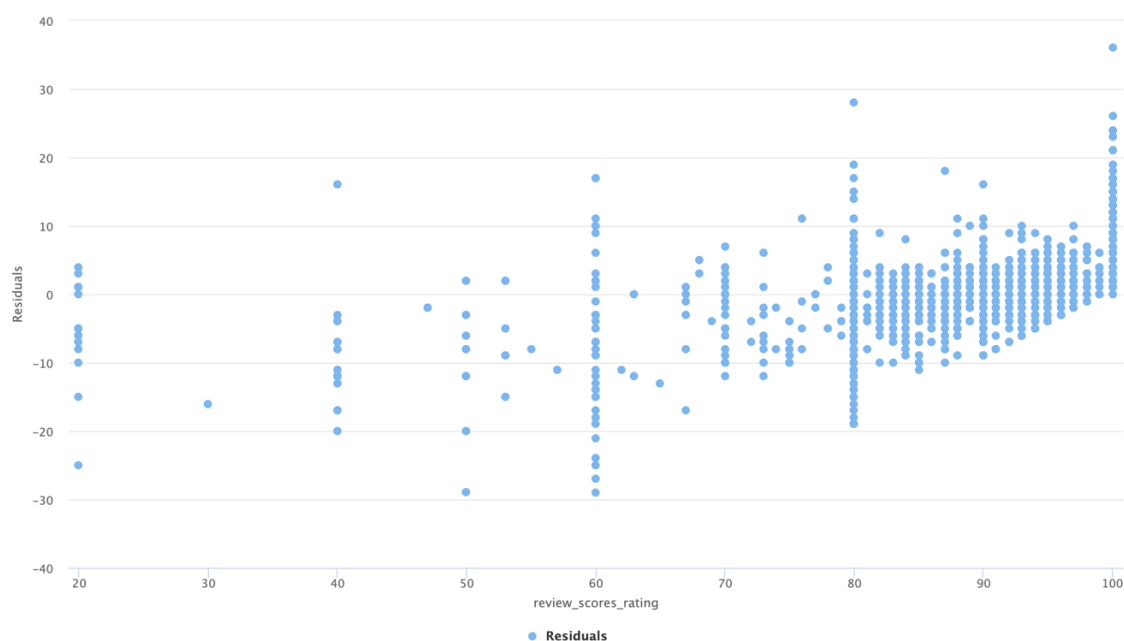


Figure 9 – Residuals plot of hold-out method

The residuals plot shows some variance in residuals as the observed values increase, particularly for high ratings near 100, where the model underpredicts high values. The model tends to perform better for lower and moderate review scores, with residuals more tightly clustered in those regions. This suggests that the model could benefit from further refinement or possibly additional predictors to better capture these ratings.

A cross-validation multiple linear regression is also developed following similar initial process as the hold-out multiple linear regression model. In cross-validation, the data is divided into 10 folds. The model is trained on k-1 folds and tested on the remaining fold. This process is repeated for all folds, and the average performance is calculated and compared with hold-out method as below.

Methods	RMSE	Squared correlation
Hold-out	4.386	0.804
Cross-validation	4.625 ± 0.183	0.780 ± 0.014

Figure 10 – Hold-out and cross-validation method evaluation metrics comparison

- RMSE of 4.625 indicates that, on average, the model's predictions are 4.625 points away from the actual review_scores_rating. The hold-out method provides a slightly lower RMSE, which could suggest better performance on the specific test set. However, cross-validation's slightly higher RMSE is expected due to the more rigorous validation process across multiple data splits.
- Squared correlation of 0.780 indicates that 78% of the variability in the review scores is explained by the model. This is a reasonably good fit; however, the squared correlation for hold-out is slightly higher.

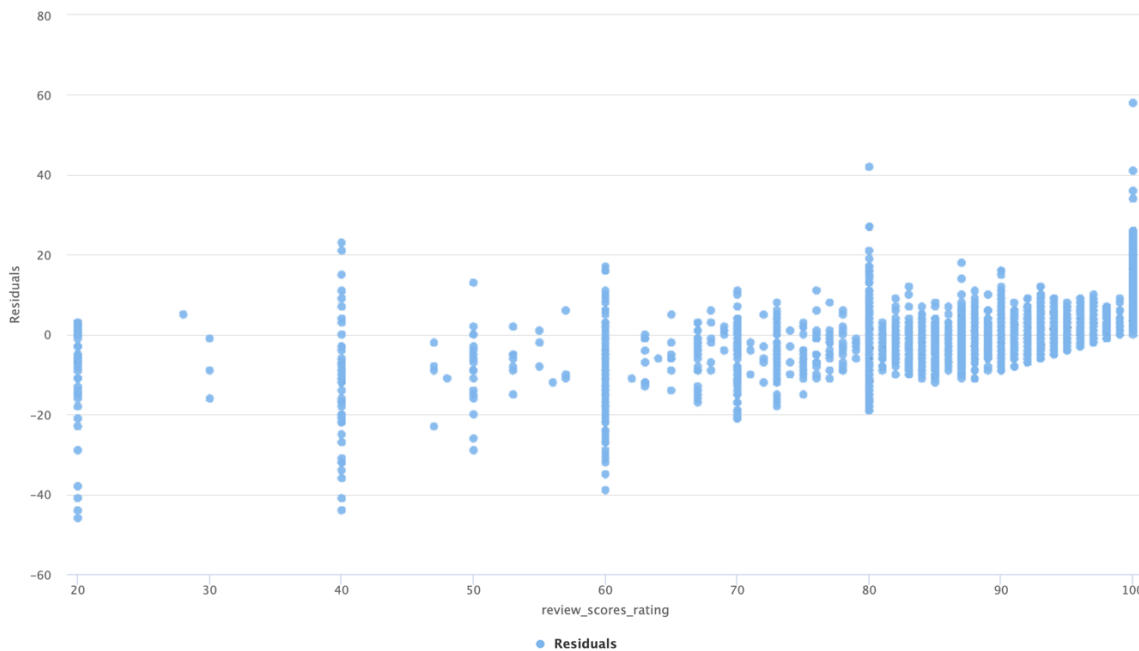


Figure 11 – Residuals plot of cross-validation method

The residuals plot also shows that the model tends to perform better for lower and moderate review scores, with residuals more tightly clustered in those regions. The model may not fully capture the attributes or patterns that lead to the highest ratings. This could be due to limitations in the feature set or the model's structure.

Generally, the hold-out model performs slightly better based on RMSE and squared correlation. Both methods show that the model struggles with higher review scores; however, cross-validation provides a more reliable performance estimate, reducing the likelihood of overfitting and making the model more applicable to unseen data.

Impact on Future Decision Making and Suggestion

The model can be used to predict review scores based on chosen features. The relatively high squared correlation and low RMSE values indicate that these features provide a solid foundation for predicting review scores. However, the model's struggle with high-score predictions suggests that additional features or refinements are needed before the model can be used in decision-making. To enhance the performance of the model, non-linear regression models should be considered. Moreover, categorical attributes can be included as predictors through proper encoding techniques.